

The Ethical Hacker's Command Handbook

Chapter 2 – Nmap (Network Mapper)

Practical Command Reference

Nmap is one of the first tools used during an authorized security assessment. This chapter focuses on the commands you are most likely to use, when to use them, and what to run next.

nmap

Purpose: Basic scan

What it does: Discover common TCP ports.

When to use: First look at a target.

Avoid when: Need only host discovery.

Next command: -sV

Memory Tip: Learn where this command fits in your workflow, not just its syntax.

-sn

Purpose: Ping scan

What it does: Check whether hosts are online without scanning ports.

When to use: Finding live hosts on a subnet.

Avoid when: Need open ports.

Next command: -sS

Memory Tip: Learn where this command fits in your workflow, not just its syntax.

-Pn

Purpose: Skip ping

What it does: Treat the host as online.

When to use: Target blocks ICMP (common in labs).

Avoid when: Large networks (slower).

Next command: -sS

Memory Tip: Learn where this command fits in your workflow, not just its syntax.

-sS

Purpose: TCP SYN scan

What it does: Fast half-open TCP scan.

When to use: Default TCP enumeration.

Avoid when: No admin/root privileges.

Next command: -sV

Memory Tip: Learn where this command fits in your workflow, not just its syntax.

-sT

Purpose: TCP Connect scan

What it does: Full TCP connect.

When to use: When SYN scan isn't available.

Avoid when: Root access is available.

Next command: -sV

Memory Tip: Learn where this command fits in your workflow, not just its syntax.

-sU

Purpose: UDP scan

What it does: Scan UDP ports.

When to use: DNS, SNMP, NTP checks.

Avoid when: Only TCP services.

Next command: -sV

Memory Tip: Learn where this command fits in your workflow, not just its syntax.

-sV

Purpose: Version detection

What it does: Identify service versions.

When to use: Research vulnerabilities.

Avoid when: Need only host discovery.

Next command: Searchsploit

Memory Tip: Learn where this command fits in your workflow, not just its syntax.

-sC

Purpose: Default NSE scripts

What it does: Run safe default scripts.

When to use: Initial service enumeration.

Avoid when: Need custom scripts only.

Next command: --script

Memory Tip: Learn where this command fits in your workflow, not just its syntax.

--script vuln

Purpose: Vulnerability scripts

What it does: Run vulnerability checks.

When to use: After identifying services.

Avoid when: Very noisy environments.

Next command: Searchsploit

Memory Tip: Learn where this command fits in your workflow, not just its syntax.

-O

Purpose: OS detection

What it does: Guess operating system.

When to use: Need OS fingerprint.

Avoid when: Firewall blocks fingerprints.

Next command: -A

Memory Tip: Learn where this command fits in your workflow, not just its syntax.

-A

Purpose: Aggressive scan

What it does: Enable OS, version, scripts, traceroute.

When to use: Small lab machines.

Avoid when: Large networks.

Next command: Save output

Memory Tip: Learn where this command fits in your workflow, not just its syntax.

-p-

Purpose: All TCP ports

What it does: Scan every TCP port.

When to use: Thorough assessment.

Avoid when: Need quick top-port scan.

Next command: -sV

Memory Tip: Learn where this command fits in your workflow, not just its syntax.

--top-ports 100

Purpose: Top ports

What it does: Scan most common ports.

When to use: Quick reconnaissance.

Avoid when: Need complete coverage.

Next command: -sV

Memory Tip: Learn where this command fits in your workflow, not just its syntax.

-T4

Purpose: Faster timing

What it does: Speed up scanning.

When to use: Typical lab environments.

Avoid when: Unstable/slow networks.

Next command: Save output

Memory Tip: Learn where this command fits in your workflow, not just its syntax.

-oA scan

Purpose: Save all formats

What it does: Normal, XML and grepable output.

When to use: Every serious assessment.

Avoid when: One-off experiments.

Next command: Review results

Memory Tip: Learn where this command fits in your workflow, not just its syntax.

Quick Workflow

1. Host discovery (-sn or -Pn if ping is blocked)
2. TCP scan (-sS)
3. Version detection (-sV)
4. Default scripts (-sC)
5. Research findings
6. Save results (-oA)

One-Page Revision

- sn Find hosts
- Pn Ignore ping
- sS Main TCP scan
- sV Versions
- sC Default scripts
- O OS detection
- A Aggressive scan
- p- All ports
- T4 Faster scan
- oA Save results